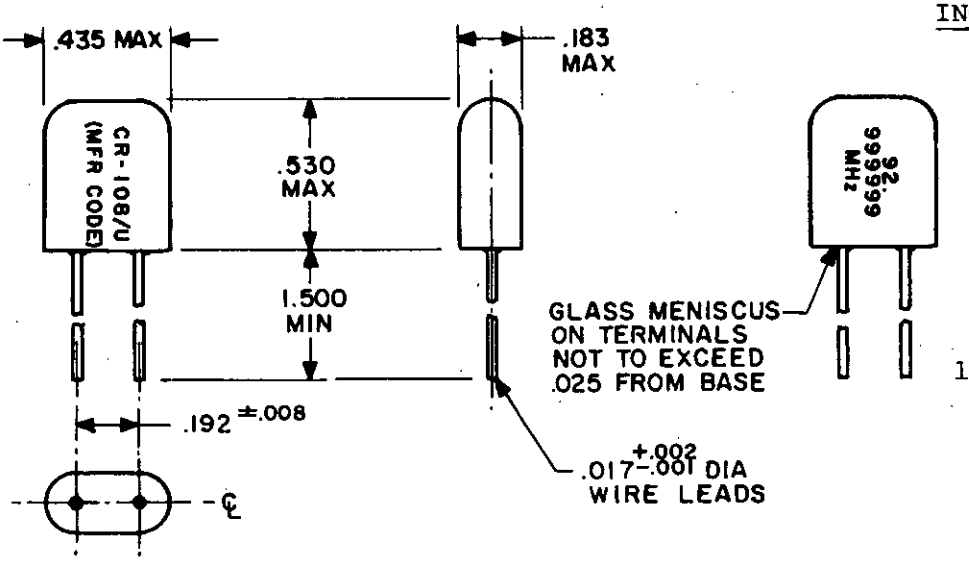
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-108/U

This specification sheet is approved for use by the Air Force Acquisition Logistics Division, Directorate of Electronic Support (AFALD/PTS), Wright-Patterson AFB, Ohio, 45433, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit, quartz, described herein shall consist of this document and the latest issue of MIL-C-3098.

Pertinent characteristics: 50 to 93 MHz; fifth mechanical overtone; controlled; series resonance.



INCHES	MM
.001	.03
.002	.05
.008	.20
.017	.43
.025	.64
.027	.69
.038	.97
.183	4.65
.192	4.88
.435	11.05
.530	13.46
1.500	38.10

NOTE:
1. Dimensions are in inches
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch equals 25.4 mm.
3. The wire leads shall be tinned, suitable for soldering.
4. The type designator, frequency and manufacturer's code shall be not less than .038 (1.97 mm) in height and not less than .027 (.69 mm) in width.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:44Z
Check the source to verify that this is the current version before use.

TABLE I. Requirements

Requirement	Specified Value or Condition
Holder	HC-26/U (MIL-H-10056/13)
Dimensions, marking and configuration.	See Figure
Frequency range.	50 to 93 MHz, inclusive
Capacitance, shunt	5.5 pF, maximum
Mode of oscillation.	Fifth mechanical overtone
Government reference standard test set	TS-683/TSM
Rated drive level.	2.0 ±0.4 mW
Calibration values	
Equivalent resistance.	40 Ohms
Resistor voltage drop.	0.28 Volt
Resonance.	Series
Reference temperature.	85° ±1°C
Operating temperature range (controlled)	80°C to 90°C, inclusive
Frequency tolerance.	±10 PPM
Equivalent resistance.	50 Ohms, maximum
Frequency stability.	±2.5 PPM
Unwanted modes 1/.	Method II
Operable temperature range	-55°C to +80°C and +90°C to +105°C, inclusive
Shock:	
Frequency change permitted	± 5 PPM
Equivalent resistance change permitted	±10 percent
Vibration: 2/	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Temperature cycling:	
Frequency change permitted	±5 PPM
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	3 PPM

1/ The minimum unwanted mode effective resistance shall be at least twice that of the main mode, or 50 Ohms, whichever is greater.

2/ Method 204, MIL-STD-202, Test Condition A.

NOTE: These crystal units were previously fabricated using Air Force Technical Order 12-1-20 and/or Collins Drawing Numbers 289-2844 through 2858-00.

Changes are not identified with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 99

User activity:
Air Force - 17

Preparing activity:
Air Force - 85

(Project No. 5955-F184-4)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:44Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-C-3098/84B(USAF)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 PREVIOUS EDITION WILL BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T16:44Z
Check the source to verify that this is the current version before use.